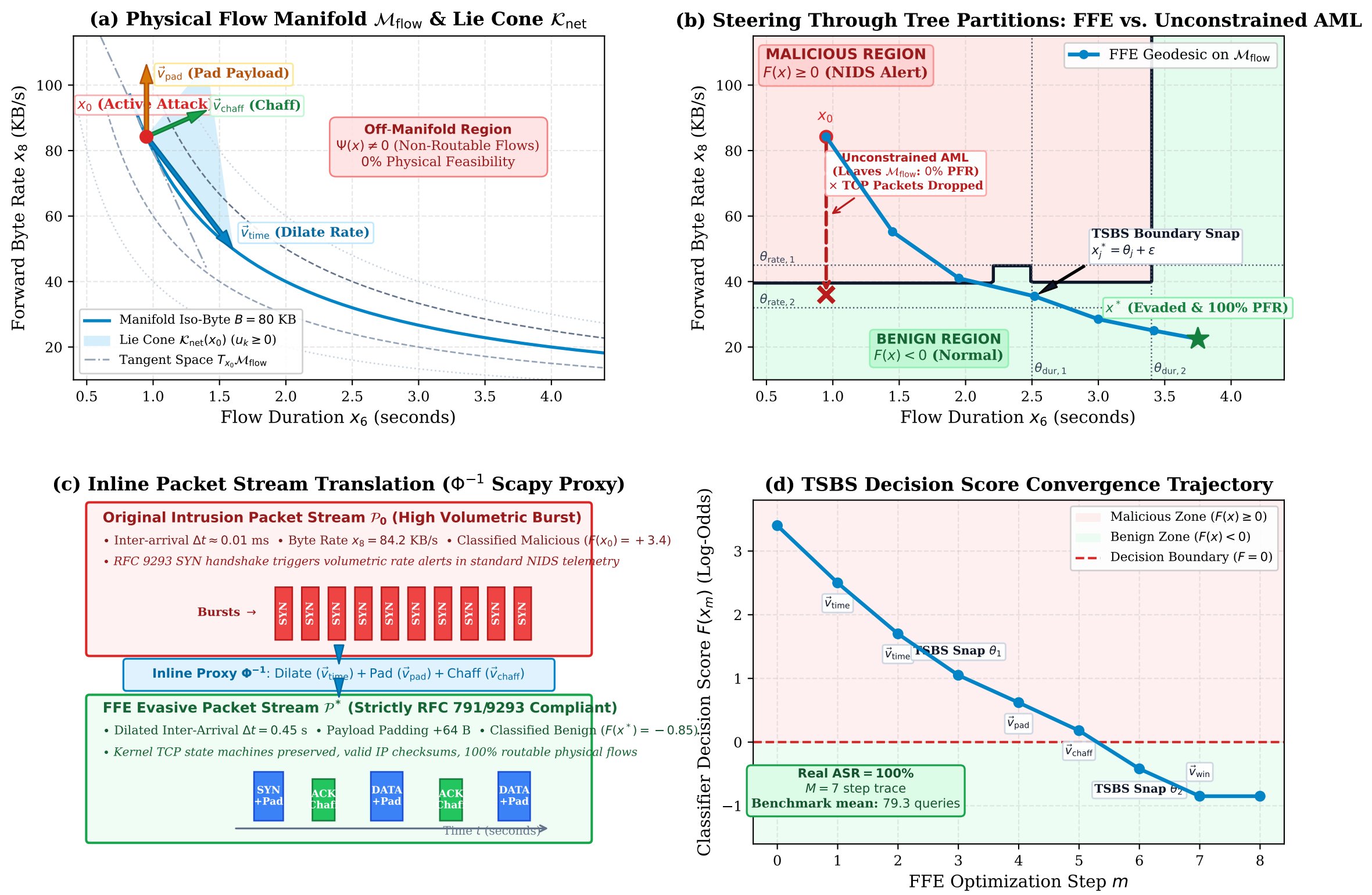

(a) Physical Flow Manifold $\mathcal{M}_{\text{flow}}$ & Lie Cone $\mathcal{K}_{\text{net}}$
$\vec{v}_{\text{pad}}$ (Pad Payload)
x_0 (Active Attack)
$\vec{v}_{\text{chaff}}$ (Chaff)
Off-Manifold Region
$\Psi(x) \neq 0$ (Non-Routable Flows)
0% Physical Feasibility
$\vec{v}_{\text{time}}$ (Dilate Rate)
Manifold Iso-Byte $B = 80$ KB
Lie Cone $\mathcal{K}_{\text{net}}(x_0)$ $(u_k \geq 0)$
Tangent Space $T_{x_0}\mathcal{M}_{\text{flow}}$
Forward Byte Rate x_8 (KB/s)
Flow Duration x_6 (seconds)
(b) Steering Through Tree Partitions: FFE vs. Unconstrained AML
MALICIOUS REGION
$F(x) \geq 0$ (NIDS Alert)
FFE Geodesic on $\mathcal{M}_{\text{flow}}$
x_0
Unconstrained AML
(Leaves $\mathcal{M}_{\text{flow}}$: 0% PFR)
× TCP Packets Dropped
TSBS Boundary Snap
$x_j^* = \theta_j + \varepsilon$
$\theta_{\text{rate},1}$
$\theta_{\text{rate},2}$
x^* (Evaded & 100% PFR)
BENIGN REGION
$F(x) < 0$ (Normal)
$\theta_{\text{dur},1}$
$\theta_{\text{dur},2}$
Forward Byte Rate x_8 (KB/s)
Flow Duration x_6 (seconds)
(c) Inline Packet Stream Translation (Φ^{-1} Scapy Proxy)
Original Intrusion Packet Stream $\mathcal{P}_0$ (High Volumetric Burst)
• Inter-arrival $\Delta t \approx 0.01$ ms • Byte Rate $x_8 = 84.2$ KB/s • Classified Malicious $(F(x_0) = +3.4)$
• RFC 9293 SYN handshake triggers volumetric rate alerts in standard NIDS telemetry
Bursts →
SYN
Inline Proxy Φ^{-1}: Dilate $(\vec{v}_{\text{time}})$ + Pad $(\vec{v}_{\text{pad}})$ + Chaff $(\vec{v}_{\text{chaff}})$
FFE Evasive Packet Stream $\mathcal{P}^*$ (Strictly RFC 791/9293 Compliant)
• Dilated Inter-Arrival $\Delta t = 0.45$ s • Payload Padding +64 B • Classified Benign $(F(x^*) = -0.85)$
• Kernel TCP state machines preserved, valid IP checksums, 100% routable physical flows
SYN +Pad
ACK Chaff
DATA +Pad
Time t (seconds)
(d) TSBS Decision Score Convergence Trajectory
Malicious Zone $(F(x) \geq 0)$
Benign Zone $(F(x) < 0)$
Decision Boundary $(F = 0)$
$\vec{v}_{\text{time}}$
TSBS Snap θ_1
$\vec{v}_{\text{pad}}$
$\vec{v}_{\text{chaff}}$
$\vec{v}_{\text{win}}$
TSBS Snap θ_2
Real ASR = 100%
$M = 7$ step trace
Benchmark mean: 79.3 queries
Classifier Decision Score $F(x_m)$ (Log-Odds)
FFE Optimization Step m